

Series Tests for Convergence and Divergence

Divergence Test

When to Use	How to Use	Conclusions	Notes
If we can see that the $\lim_{n \rightarrow \infty} a_n \neq 0$, use the divergence test	$\lim_{n \rightarrow \infty} a_n$	If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ diverges If $\lim_{n \rightarrow \infty} a_n = 0$ the test is inconclusive	1) This test does not show convergence 2) This can be used with an alternating series

Geometric Series Test

When to Use	Conclusions	Notes
Use Geometric Series test if it is in the form: $\sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n$	The series converges to $\frac{a}{1-r}$ if $ r < 1$ The series diverges if: $ r \geq 1$	1) Useful if n is only in the exponent. Simplification may be needed 2) This is the ONLY test that tells us what a series converges to.

Integral Test

When to Use	Conditions	Conclusions	Notes
Use the Integral Test if $\int_k^{\infty} f(x) dx$ is easily evaluated	Suppose $f(x)$ is: 1) continuous on $[k, \infty)$ 2) positive on $[k, \infty)$ 3) ultimately decreasing 4) $a_n = f(n)$	If $\int_k^{\infty} b_n$ converges, then a_n converges If $\int_k^{\infty} b_n$ diverges, then a_n diverges	1) This test is never inconclusive. 2) The value of the integral is NOT the value of the sum.

P-Series Test

When to Use	Conclusions	Notes
If the denominator is raised to a constant denominator p , use the P-Series test.	If $p > 1$ the series converges If $p \leq 1$ the series diverges	1) Usually in the form $\sum \frac{1}{n^p}$ 2) Can be used for comparison tests

Alternating Series Test

When to Use	Conditions	Conclusions
Has the form $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ Will converge if all three conditions are met.	1) a_n 's are all positive 2) Terms are decreasing, $a_n \geq a_{n+1}$ for all n 3) $\lim_{n \rightarrow \infty} a_n = 0$	If any of these conditions are not met the test is inconclusive. If they are all met, the series converges.

Alternating Series Helps:

- 1) Before testing, pull out all of the $(-1)^n$ terms so that all the remaining terms are positive.
- 2) This test is only concerned with a_n and not with $(-1)^n a_n$.
- 3) This test has a remainder estimation formula:

$$R = \left| \sum_{n=1}^{\infty} (-1)^{n+1} a_n - \sum_{n=1}^N (-1)^{n+1} a_n \right| \leq a_{N+1}$$

Absolute Convergence:

A series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges.
If $\sum a_n$ converges and $\sum |a_n|$ diverges, the $\sum a_n$ converges conditionally.

Comparison Test

When to Use	Conditions	Conclusions	Notes
Use when a series has a similar form to a p-series or a geometric series	Let $\sum a_n$ and $\sum b_n$ be series with positive terms.	If $\sum a_n \leq \sum b_n$ for all n sufficiently large, then $\sum a_n$ converges If $\sum a_n \geq \sum b_n$ for all n sufficiently large, then $\sum a_n$ diverges	1) Use to compare a complicated series with a less-complicated series that grows at the same rate. 2) Need to come up with own series b_n

Limit Comparison Test

When to Use	Conditions	Conclusions
Use to show that the series $\sum a_n$ and $\sum b_n$ are growing at the same rate.	Let $\sum a_n$ and $\sum b_n$ be series with positive terms. If series are decreasing, we can test for absolute convergence, $\sum a_n$.	If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ where $c > 0$ both series diverge or converge. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, $\sum a_n$ converges. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, $\sum a_n$ diverges.

Ratio Test

When to Use	How to Use	Conclusions
Use when we have n 's in the exponents and connected to constants. This is also useful for factorials ($n!$)	$\lim_{n \rightarrow \infty} \left \frac{a_{n+1}}{a_n} \right = p$	If $p < 1$, then $\sum a_n$ converges absolutely. If $p > 1$, then $\sum a_n$ diverges. If $p = 1$, then the test is inconclusive.

Root Test

When to Use	How to Use	Conclusions
Useful if the series has a power of n . This test is not commonly used.	$\lim_{n \rightarrow \infty} \left \sqrt[n]{ a_n } \right = p$	If $p < 1$, then $\sum a_n$ converges absolutely. If $p > 1$, then $\sum a_n$ diverges. If $p = 1$, then the test is inconclusive.

Commonly Occurring Limits

- $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$
- $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$
- $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n \rightarrow \infty} x^n = 0 \quad (|x| < 1)$
- $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$